

Time Complexity of Binary Search Tree

Commonly performed operations on binary search tree are-



1. Search Operation
2. Insertion Operation
3. Deletion Operation

In this article, we will discuss time complexity of BST Operations.

Time Complexity-

- Time complexity of all BST Operations = $O(h)$.
- Here, h = Height of binary search tree

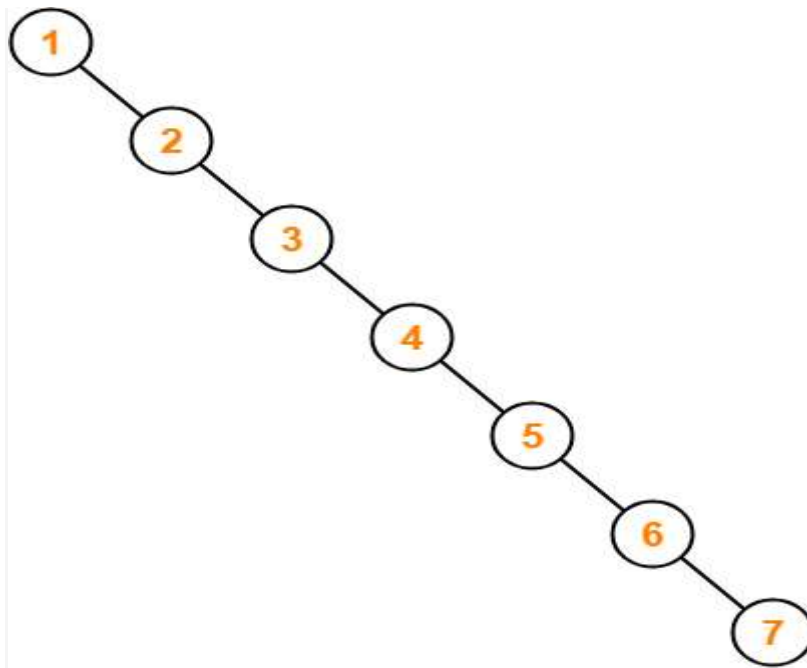
Now, let us discuss the worst case and best case.

Worst Case-

In worst case,

- The binary search tree is a skewed binary search tree.
- Height of the binary search tree becomes n .
- So, Time complexity of BST Operations = $O(n)$.

In this case, binary search tree is as good as unordered list with no benefits.

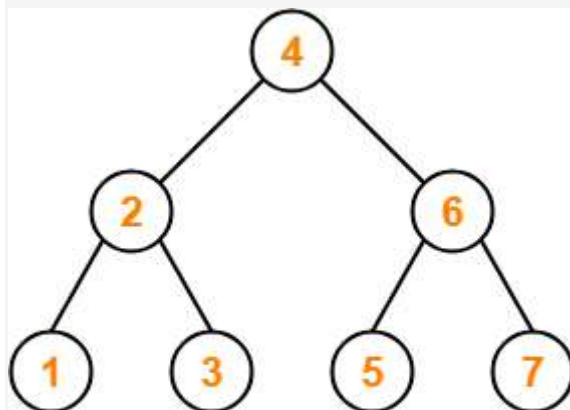


Skewed Binary Search Tree

Best Case-

In best case,

- The binary search tree is a balanced binary search tree.
- Height of the binary search tree becomes $\log(n)$.
- So, Time complexity of BST Operations = $O(\log n)$.



Balanced Binary Search Tree